ARYA LAKSMANA DEWANATA

+62 857 0153 1222 | arlaksde@gmail.com

https://arlaksde.github.io/portfolio/portfolio-main

Arya Laksmana Dewanata | LinkedIn

SUMMARY

Final-year Electronics Engineering Technology student (D4 / equivalent to a Bachelor's degree) with a solid 7-year technical background in electronics. I'm certified in Advantech IIoT & Edge Computing and have hands-on experience in developing production monitoring systems (OEE), working with industrial protocols (PLC Omron, Haiwell, Modbus, RS485), and lab experience in instrumentation, fault analysis, and advanced control systems (PID, Fuzzy Logic).

INTERNSHIP

January – March 2021

IoT Embedded System Intern, PPTIK ITB — online internship

- Monitoring System Integration: Calibrating sensors and programming automated control systems to ensure data accuracy in real-time soil moisture management.
- Personnel Management Digitalization: Developing an RFID-based attendance system architecture to improve data validity and efficiency of attendance management within the organization.

June 2021 — August 2021

Manufacturing Process Intern, Panasonic Manufacturing Indonesia — Jl. Raya Bogor KM 29, Gandaria, Jakarta

- Production Line Optimization: Responsible for overseeing and implementing the aluminum coating process on technical components in the refrigeration department in accordance with global manufacturing quality standards.
- Operational Efficiency: Collaborates within the technical team to achieve daily production targets through the application of continuous improvement principles.

March 2026 – June 2026

Internship Engineering Electronic, Ruang Industri Indonesia — Jalan Mulawarman Selatan 50278 Tembalang Central Java

- Designed and implemented an end-to-end Overall Equipment Effectiveness (OEE) monitoring system utilizing an Orange Pi single-board computer as the central on-premise server.
- Assisted in assembling and programming RIKUB (Riset Konsorium Unggulan Berdampak)

EDUCATION

Sep 2022 – Sep 2026 (Expected)

Politeknik Negeri Semarang (GPA 3.6) — Jalan Prof. H. Soedarto, Tembalang, Kota Semarang, Jawa Tengah

D4 Teknologi Rekayasa Elektronika (S1 Terapan / Equivalent to Bachelor's Degree)

- **Control Systems & Robotics:** Programmed line follower robots and simulated autonomous systems using Webots. Implemented PID controllers for stepper motors and managed encoder data processing using Matlab.
- **Applied Electronics:** Designed, calculated, and fabricated various analog and digital circuits, including precision heat control systems, and water level using Arduino ide.

PROJECTS

May – July 2026	**Riset Konsorsium Unggulan Berdampak (RIKUB)** Assisted in assembling and programming RIKUB (C/C++ Arduino as a Driver Stepper for Motor Nema 23 and Waterpump Peristaltic	Ruang Industri Indonesia
March – June 2026	**Overall Equipment Effectiveness** Designed and implemented an end-to-end Overall Equipment Effectiveness (OEE) monitoring system utilizing an Orange Pi single-board computer as the central on-premise server.	Ruang Industri Indonesia
June 2025 – January 2026	**Lecturer Attendance and Appointment Information System Based on Face Recognition** Engineered an integrated appointment and availability management system for lecturers and students. By combining face recognition technology, IoT devices, and a web application, this system significantly streamlines scheduling and enhances academic communication efficiency.	Politeknik Negeri Semarang
September 2025 – February 2026	**Smart Aquaculture System** Developed an IoT-based automated maintenance system using an ESP32 microcontroller to ensure a stable and optimal aquatic ecosystem. The system continuously monitors crucial water quality metrics via pH, TDS, and water level sensors, while seamlessly executing scheduled automated draining.	Freelance Project
January – March 2022	**Stamping Machine with Pneumatic Integrated with PLC Omron** Creating a prototype of an automatic stamping machine that integrates pneumatic actuators with PLC-based control logic.	SMK Negeri 1 Semarang
January – March 2021	**Soil Moisture Monitoring System** Assembling an analog circuit and configuring a soil moisture sensor network to improve real-time water management data accuracy in a precision farming system. **Online Attendance Using RFID** Developing an online attendance system architecture based on RFID technology to improve data accuracy and the efficiency of managing personnel attendance in an organization.	PPTIK ITB (Remote Project)

SKILLS

- **Languages:** English (TOEIC 570), Mandarin (HSK 3)
- **Hardware & Protocols:** PLC, SBC (Orange-Pi, Raspberry-Pi), Microcontroller (ESP32), RS485, Modbus
- **Software & Programming:** C/C++, Python, MATLAB, Flask, OpenCV (Haar Cascade), Proteus, Lazarus, CorelDRAW